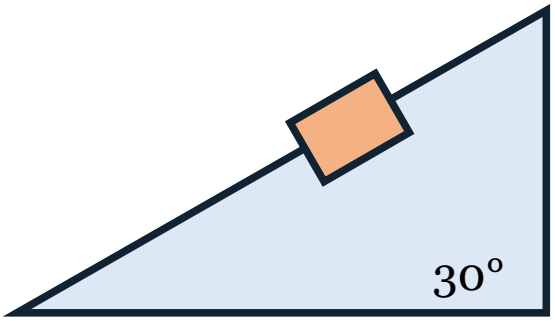
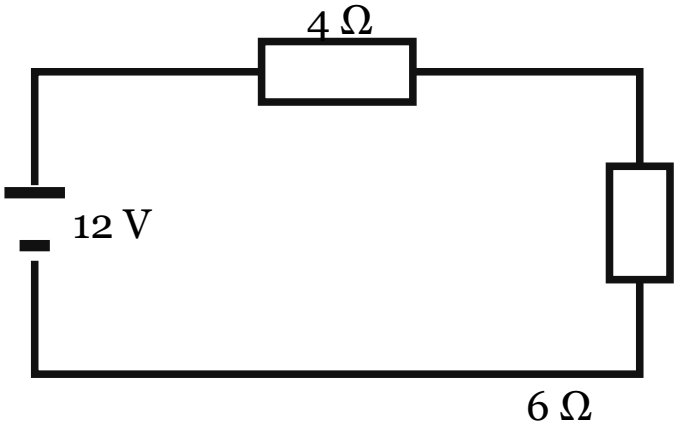
Q1. A block of mass 2 kg rests on a smooth incline of 30° as shown in the figure. Find its acceleration.

Fig. 1

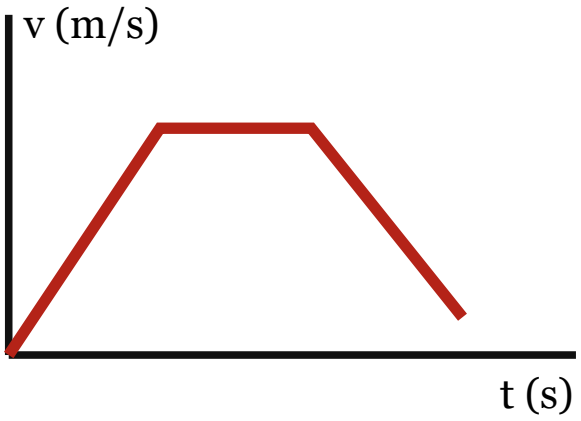


Q2. The dimensional formula of Planck's constant is (A) ML^2T^{-1} (B) MLT^{-2} (C) ML^2T^{-2} (D) MLT^{-1}

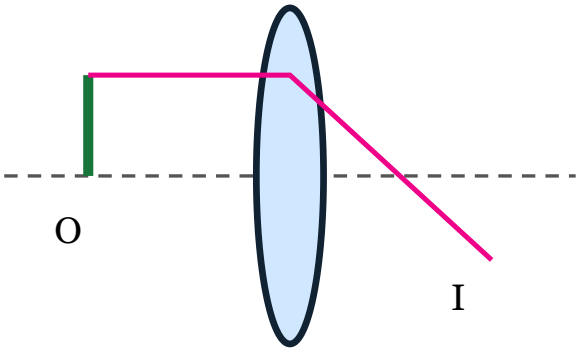
Q3. In the circuit shown, find the current through the 4 Ω resistor.



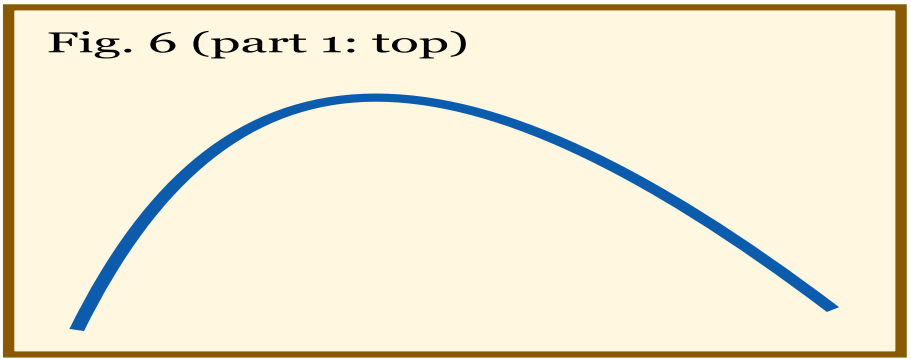
Q4. The velocity–time graph of a particle is shown. Find the distance travelled.



Q5. An object is placed in front of a convex lens as shown. Locate the image.



Q6. The curve below continues on the next page. Find the maximum.



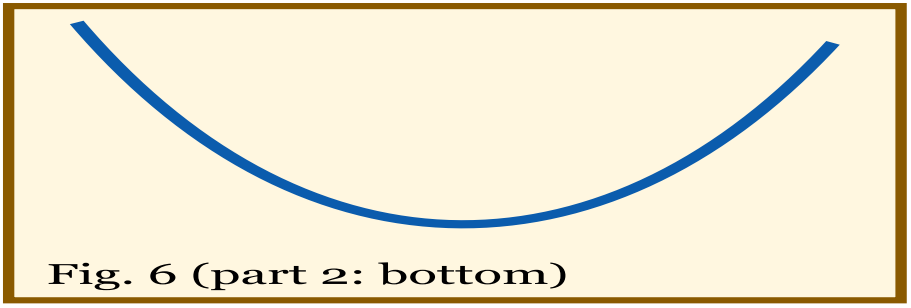
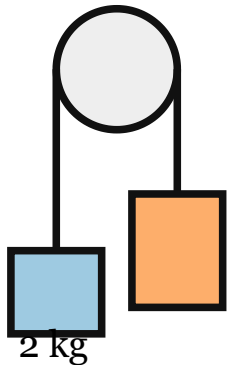
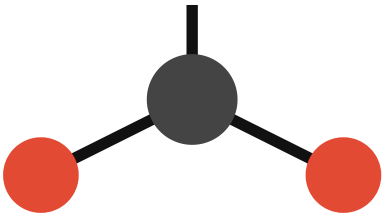
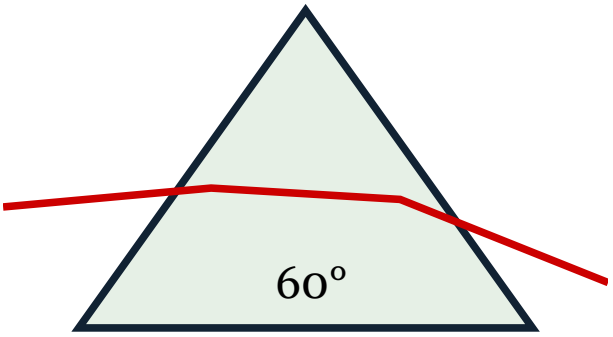


Fig. 6 (part 2: bottom)

Q7. Two arrangements are shown: (a) a pulley system and (b) a molecule.





Structure X

Q8. A ray passes through the prism shown. Find the angle of deviation.